INFO8006 Introduction to Artificial Intelligence

Exercise session 5: Reasoning over time

Recap

Markov models

The world is a discrete series of time slices, where $\mathbf{X}_t$ denotes the set of *unobservable* state variables and $\mathbf{E}_t$ the set of *observable* evidence variables at time t. We specify a prior $\mathbf{P}(\mathbf{X}_0)$, a *transition model* $\mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{0:t-1})$ and a *sensor model* $\mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_{0:t}, \mathbf{E}_{0:t-1})$, under

- the first-order *Markov assumption* $\mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{0:t-1}) = \mathbf{P}(\mathbf{X}_t \mid \mathbf{X}_{t-1})$, which defines a first-order *Markov process*;
- the *sensor Markov assumption* $\mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_{0:t}, \mathbf{E}_{0:t-1}) = \mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_t)$;
- the *stationarity assumption*: the transition and sensor models are the same for all t.

The joint distribution of all variables up to t is then

$$\mathbf{P}(\mathbf{X}_{0:t}, \mathbf{E}_{1:t}) = \mathbf{P}(\mathbf{X}_0) \prod_{i=1}^{t} \mathbf{P}(\mathbf{X}_i \mid \mathbf{X}_{i-1}) \mathbf{P}(\mathbf{E}_i \mid \mathbf{X}_i).$$

Unrolled over time as a Bayesian network, the same slice of variables and arrows repeated at every t, the model is a *dynamic Bayesian network*; a slice may hold several state and evidence variables.

Hidden Markov models

A *hidden Markov model* (HMM) is a Markov process in which the state $\mathbf{X}_t = X_t$ and the evidence $\mathbf{E}_t = E_t$ are both single discrete random variables, with domains $D_{X_t} = \{1, \ldots, S\}$ and $D_{E_t} = \{1, \ldots, R\}$. Its models become matrices:

- the prior $\mathbf{P}(X_0)$ becomes a column vector $\mathbf{f}_0 \in \mathbb{R}_+^S$;
- the transition model $\mathbf{P}(X_t \mid X_{t-1})$ becomes an $S \times S$ *transition matrix* $\mathbf{T}$ such that $\mathbf{T}_{ij} = P(X_t = j \mid X_{t-1} = i)$;
- the sensor model $\mathbf{P}(E_t \mid X_t)$ is an $S \times R$ *sensor matrix* $\mathbf{B}$ such that $\mathbf{B}_{ij} = P(E_t = j \mid X_t = i)$;
- the *observation matrix* $\mathbf{O}_t$ is the diagonal matrix of the column e_t of $\mathbf{B}$, $(\mathbf{O}_t)_{jj} = \mathbf{B}_{je_t}$.

Several evidence variables, conditionally independent given the state, merge into one evidence variable whose values are their combinations, with probabilities the products of theirs. In index notation, $(\mathbf{MN})_{ij} = \sum_k \mathbf{M}_{ik}\mathbf{N}_{kj}$ and $(\mathbf{T}^T\mathbf{f})_j = \sum_i \mathbf{T}_{ij}\mathbf{f}_i$.

Inference tasks

Given the evidence $\mathbf{e}_{1:t}$ collected up to time t, we define

- *filtering*: $\mathbf{P}(\mathbf{X}_t \mid \mathbf{e}_{1:t})$, the *belief state* about the current hidden state $\mathbf{X}_t$;
- *prediction*: $\mathbf{P}(\mathbf{X}_{t+k} \mid \mathbf{e}_{1:t})$ for $k > 0$;
- *smoothing*: $\mathbf{P}(\mathbf{X}_k \mid \mathbf{e}_{1:t})$ for $0 \leq k < t$;
- *most likely explanation*: $\arg\max_{\mathbf{x}_{1:t}} P(\mathbf{x}_{1:t} \mid \mathbf{e}_{1:t})$.

Each is computed by a recursion over t, below. The proportionality signs hide a normalization constant, written $\alpha = 1/Z$.

Filtering and prediction

The *Bayes filter* maintains the belief state, or *forward message*, $\mathbf{f}_{1:t} = \mathbf{P}(\mathbf{X}_t \mid \mathbf{e}_{1:t})$, by alternating two steps:

$$\text{predict:} \qquad \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t}) = \sum_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t) P(\mathbf{x}_t \mid \mathbf{e}_{1:t}),$$

$$\text{update:} \qquad \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t+1}) \propto \mathbf{P}(\mathbf{e}_{t+1} \mid \mathbf{X}_{t+1}) \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t}),$$

with $Z = P(\mathbf{e}_{t+1} \mid \mathbf{e}_{1:t})$. *Prediction* repeats the predict step up to $t + k$, without update. For an HMM, $\mathbf{f}_{1:t+1} = \alpha\, \mathbf{O}_{t+1} \mathbf{T}^T \mathbf{f}_{1:t}$. For continuous variables, sums are replaced with integrals and probability mass functions with densities:

$$p(\mathbf{x}_{t+1} \mid \mathbf{e}_{1:t+1}) \propto p(\mathbf{e}_{t+1} \mid \mathbf{x}_{t+1}) \int p(\mathbf{x}_{t+1} \mid \mathbf{x}_t) p(\mathbf{x}_t \mid \mathbf{e}_{1:t})\, \mathrm{d}\mathbf{x}_t.$$

Stationary distribution (optional)

Without new evidence, the predicted distribution may converge, as k grows, to a *stationary distribution*: a fixed point $\mathbf{P}(\mathbf{X}_{t+1}) = \mathbf{P}(\mathbf{X}_t)$ of the predict step. For an HMM, it is a vector $\mathbf{f}$ such that $\mathbf{f} = \mathbf{T}^T \mathbf{f}$.

Smoothing

Dividing the evidence $\mathbf{e}_{1:t}$ into $\mathbf{e}_{1:k}$ and $\mathbf{e}_{k+1:t}$,

$$\mathbf{P}(\mathbf{X}_k \mid \mathbf{e}_{1:t}) \propto \mathbf{P}(\mathbf{X}_k \mid \mathbf{e}_{1:k}) \mathbf{P}(\mathbf{e}_{k+1:t} \mid \mathbf{X}_k) = \alpha\, \mathbf{f}_{1:k} \times \mathbf{b}_{k+1:t},$$

where $\times$ is a pointwise multiplication of vectors and the *backward message* $\mathbf{b}_{k+1:t} = \mathbf{P}(\mathbf{e}_{k+1:t} \mid \mathbf{X}_k)$ is computed by backwards recursion, $\mathbf{b}_{k+1:t} = \text{backward}(\mathbf{b}_{k+2:t}, \mathbf{e}_{k+1})$. For an HMM, $\mathbf{b}_{k+1:t} = \mathbf{T}\mathbf{O}_{k+1}\mathbf{b}_{k+2:t}$, from the all-one vector $\mathbf{b}_{t+1:t}$ of size S.

Most likely explanation (optional)

The *Viterbi algorithm* replaces the forward message with $\mathbf{m}_{1:t} = \max_{\mathbf{x}_{1:t-1}} \mathbf{P}(\mathbf{x}_{1:t-1}, \mathbf{X}_t \mid \mathbf{e}_{1:t})$, the probability of the most likely path to each state, computed from $\mathbf{m}_{1:1} = \mathbf{f}_{1:1}$ as

$$\mathbf{m}_{1:t+1} \propto \mathbf{P}(\mathbf{e}_{t+1} \mid \mathbf{X}_{t+1}) \max_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t) \mathbf{m}_{1:t}.$$

For each value $\mathbf{x}_{t+1}$, it records the maximizing predecessor $\psi_{t+1}(\mathbf{x}_{t+1}) = \arg\max_{\mathbf{x}_t} P(\mathbf{x}_{t+1} \mid \mathbf{x}_t) \mathbf{m}_{1:t}(\mathbf{x}_t)$. The most likely path ends in $\mathbf{x}_t^* = \arg\max_{\mathbf{x}_t} \mathbf{m}_{1:t}(\mathbf{x}_t)$ and is recovered backwards, $\mathbf{x}_k^* = \psi_{k+1}(\mathbf{x}_{k+1}^*)$.

Cheat sheet for Gaussian distributions

If $\mathbf{x}$ and $\mathbf{y}$ have the joint Gaussian distribution

$$p\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} = \mathcal{N}\left(\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} \middle| \begin{pmatrix}\mathbf{a}\\ \mathbf{b}\end{pmatrix}, \begin{pmatrix}\mathbf{A} & \mathbf{C}\\ \mathbf{C}^T & \mathbf{B}\end{pmatrix}\right),$$

then the marginal and conditional distributions of $\mathbf{x}$ and $\mathbf{y}$ are given by the following identities, whose letters are generic (they are not the matrices of the HMM or of the Kalman filter):

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x} \mid \mathbf{a}, \mathbf{A}),\\ p(\mathbf{y}) &= \mathcal{N}(\mathbf{y} \mid \mathbf{b}, \mathbf{B}),\\ p(\mathbf{x} \mid \mathbf{y}) &= \mathcal{N}(\mathbf{x} \mid \mathbf{a} + \mathbf{C}\mathbf{B}^{-1}(\mathbf{y} - \mathbf{b}), \mathbf{A} - \mathbf{C}\mathbf{B}^{-1}\mathbf{C}^T),\\ p(\mathbf{y} \mid \mathbf{x}) &= \mathcal{N}(\mathbf{y} \mid \mathbf{b} + \mathbf{C}^T\mathbf{A}^{-1}(\mathbf{x} - \mathbf{a}), \mathbf{B} - \mathbf{C}^T\mathbf{A}^{-1}\mathbf{C}). \end{aligned}$$

If $p(\mathbf{x}) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $p(\mathbf{y} \mid \mathbf{x}) = \mathcal{N}(\mathbf{y} \mid \mathbf{H}\mathbf{x} + \mathbf{u}, \mathbf{R})$, then the joint distribution of $\mathbf{x}$ and $\mathbf{y}$ is Gaussian with

$$p\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} = \mathcal{N}\left(\begin{pmatrix}\mathbf{x}\\ \mathbf{y}\end{pmatrix} \middle| \begin{pmatrix}\boldsymbol{\mu}\\ \mathbf{H}\boldsymbol{\mu} + \mathbf{u}\end{pmatrix}, \begin{pmatrix}\boldsymbol{\Sigma} & \boldsymbol{\Sigma}\mathbf{H}^T\\ \mathbf{H}\boldsymbol{\Sigma} & \mathbf{H}\boldsymbol{\Sigma}\mathbf{H}^T + \mathbf{R}\end{pmatrix}\right).$$

Kalman filter

The *Kalman filter* is the Bayes filter for continuous state and evidence variables under a Gaussian prior $p(\mathbf{x}_0)$, a linear Gaussian transition model and a linear Gaussian sensor model,

$$\begin{aligned} p(\mathbf{x}_{t+1} \mid \mathbf{x}_t) &= \mathcal{N}(\mathbf{x}_{t+1} \mid \mathbf{A}\mathbf{x}_t + \mathbf{b}, \mathbf{Q}),\\ p(\mathbf{e}_t \mid \mathbf{x}_t) &= \mathcal{N}(\mathbf{e}_t \mid \mathbf{H}\mathbf{x}_t + \mathbf{d}, \mathbf{R}). \end{aligned}$$

Under these assumptions, the predict and update steps both yield Gaussian distributions, $p(\mathbf{x}_{t+1} \mid \mathbf{e}_{1:t}) = \mathcal{N}(\mathbf{x}_{t+1} \mid \boldsymbol{\mu}^-_{t+1}, \boldsymbol{\Sigma}^-_{t+1})$ and $p(\mathbf{x}_{t+1} \mid \mathbf{e}_{1:t+1}) = \mathcal{N}(\mathbf{x}_{t+1} \mid \boldsymbol{\mu}_{t+1}, \boldsymbol{\Sigma}_{t+1})$, whose parameters are derived in closed form with the Gaussian identities above.

Particle filter

When the models are not linear Gaussian, a *particle filter* approximates $p(\mathbf{x}_t \mid \mathbf{e}_{1:t})$ with N samples $\mathbf{x}^1_t, \ldots, \mathbf{x}^N_t$, the *particles*. Each step propagates every particle through the transition model, $\mathbf{x}^i_{t+1} \sim p(\mathbf{x}_{t+1} \mid \mathbf{x}^i_t)$, weights it by $w^i = p(\mathbf{e}_{t+1} \mid \mathbf{x}^i_{t+1})$, then resamples N particles with probabilities proportional to the weights.

In session: exercises 1 and 3 (November 5), 4 and 5 with the Kalman notebook (November 12).
Recommended before the next session: exercise 2 (d)–(e).

Exercise 1 Traffic on a road (Exam, January 2023)
A navigation app estimates the traffic on a road, minute by minute. Each minute, the road is either congested or fluid. From past data, the app knows that if the road is congested, there is a 40% chance that it is still congested the next minute. If the road is fluid, there is a 70% chance that it is still fluid the next minute. The app cannot see the traffic, but each minute it receives the speed of one car driving on the road, which is either fast, normal or slow. If the road is congested, the car is fast with probability 20% and slow with probability 60%. If the road is fluid, it is fast with probability 50% and slow with probability 30%.

(a) Define the components of a hidden Markov model (HMM), where the state $X_t \in \{\texttt{congested}, \texttt{fluid}\}$ is the traffic at minute t and the evidence $E_t \in \{\texttt{fast}, \texttt{normal}, \texttt{slow}\}$ is the speed of the car received at minute t. At the start, the app has no information about the traffic, so that the prior $\mathbf{P}(X_0)$ is uniform.

(b) At some point, the app only receives a speed every other minute (at minutes 1, 3, 5, ..., and none at minutes 2, 4, ...). Sketch a Bayesian network describing this new process.

(c) Express, in terms of the HMM components, the distribution $\mathbf{P}(X_t \mid e_{1:t:2})$ of the traffic at minute t given the speeds $e_{1:t:2}$ received every other minute. Note that $e_{1:6:2} = (e_1, e_3, e_5)$ and $e_{1:7:2} = (e_1, e_3, e_5, e_7)$. Separate the cases where t is even and odd.

(d) Given the observations $e_{1:3:2} = (e_1, e_3) = (\texttt{slow}, \texttt{fast})$, compute the probability that the road is congested at minute 3.

(e) Without any observation, which fraction of the time is the road congested in the long run?

Exercise 2 Umbrella world (AIMA ch. 14)
A security robot guards a secret underground installation. It would like to know whether it is raining today, but its only access to the outside world occurs each morning, when it sees the director coming in with, or without, an umbrella. For each day t, the evidence is a single variable $\mathit{Umbrella}_t \in \{\text{true}, \text{false}\}$, or U_t for short, *i.e.* whether the umbrella appears or not, and the (hidden) state is a single variable $\mathit{Rain}_t \in \{\text{true}, \text{false}\}$, or R_t, *i.e.* whether it is raining or not.

The robot's model assumes that from one day $t-1$ to the next t, the chances that the weather stays the same are 70 %. It also assumes that the director brings an umbrella 90 % of the time when it is raining, and 20 % of the time otherwise.

(a) The robot represents the umbrella world as a Markov model. What formal assumptions correspond to these beliefs?

(b) Sketch a Bayesian network structure describing the umbrella world and provide the transition and sensor models.

(c) Express the distributions $\mathbf{P}(R_{t+1} \mid R_{t-1})$, $\mathbf{P}(U_t \mid R_{t-1})$ and $\mathbf{P}(R_t \mid R_{t-1}, U_t)$ in terms of the transition and sensor models.

(d) Suppose the robot observes an unending sequence of days on which the umbrella appears. Show that, as the days go by, the probability of rain on the current day tends monotonically towards a fixed point. Calculate this fixed point.

(e) Now consider forecasting further and further into the future, given t umbrella observations. Is there a fixed point? If yes, compute its exact value.

Exercise 3 Coin-tossing machine

The surveillance program of a casino watches a coin-tossing machine. The machine holds 3 biased coins a, b and c, whose probabilities of heads are 80 %, 50 % and 20 %, respectively. It first picks one of the coins at random. Then, 4 times, it either keeps the current coin with probability $\frac{2}{3}$ or replaces it by one of the two other coins, chosen at random, then tosses the coin and shows the result. The program only sees the results: heads, heads, tails, heads.

(a) Provide a hidden Markov model (HMM) that describes the process.

(b) What are the probabilities of the last coin given the sequence of results?

(c) What are the probabilities of the coin picked first given the sequence of results? And of the first coin tossed?

(d) What is the most likely sequence of tossed coins? (optional)

Exercise 4 Robot state estimation (Exam, January 2024)

Let us consider a continuous first-order Markov process describing the motion of a robot in a one-dimensional world. On average, the robot moves forward by a distance b at each timestep. The robot's position x_t at time t is not directly observable, but can be estimated using a sensor that measures the position of the robot relative to the sensor. Assuming

- the sensor is fixed at position 0,
- a Gaussian prior $\mathcal{N}(x_0 \mid \mu_0, \sigma_0^2)$ on the initial position of the robot,
- a Gaussian transition model $\mathcal{N}(x_{t+1} \mid x_t + b, \sigma_x^2)$,
- a Gaussian sensor model $\mathcal{N}(e_t \mid x_t, \sigma_e^2)$,

answer the following questions.

(a) Derive the update equations of the Bayes filter. More specifically, since the belief distribution is Gaussian, derive the update equations of its mean μ_{t+1} and variance σ_{t+1}^2.

(b) Do the update equations make sense in the following limiting cases? Use your physical intuition.

- The sensor is extremely noisy.
- The sensor is perfect.
- The transition model is extremely noisy.
- The transition model is perfect.

(c) Describe one step of a particle filter for this model, from N particles $x_t^1, \ldots, x_t^N$ that represent $p(x_t \mid e_{1:t})$ to particles that represent $p(x_{t+1} \mid e_{1:t+1})$. What do the particles approximate, and when is a particle filter preferable to the update equations of (a)?

Exercise 5 Ball on a spring

The "Foire de Liège" has a new attraction called "Super Spring Ultra Pro Max XXL", which consists of a ball attached to a spring on a platform. The participants take a seat in the ball, locked at some position. The ball is then released and pulled by the spring, which makes it oscillate back and forth. After a few seconds, the ball is stopped by magnetic brakes. You notice that the final position of the ball is different each time. As the ball is opaque, you wonder if it is possible for the participants to guess where the ball stopped, given their perception of acceleration.

You more or less remember your Newtonian mechanics class and model the movement of the ball as a series of transitions

$$\begin{aligned} z_t &= z_{t-1} + \Delta t\, \dot{z}_{t-1} + \frac{1}{2}\Delta t^2\, \ddot{z}_{t-1} \\ \dot{z}_t &= \dot{z}_{t-1} + \Delta t\, \ddot{z}_{t-1} \\ \ddot{z}_t &= \rho w_{t-1} - \kappa\, z_{t-1} - \eta\, \dot{z}_{t-1} \\ w_t \mid w_{t-1} &\sim \mathcal{N}(\lambda w_{t-1}, \sigma_w^2) \end{aligned}$$

where z_t, $\dot{z}_t$ and $\ddot{z}_t$ are respectively the position, velocity and acceleration of the ball at timestep t, w_t is the wind at timestep t, Δt is the time elapsed between $t-1$ and t, ρ is the thrust coefficient of the wind, κ is the stiffness of the spring, η is the friction coefficient of the platform and λ is the persistence of the wind. You estimate that the ball starts at position $z_0 = 10.0 \pm 0.5\,\mathrm{m}$ with a negligible speed $0.0 \pm 0.1\,\mathrm{m\,s^{-1}}$ and acceleration $0.0 \pm 0.1\,\mathrm{m\,s^{-2}}$. You assume that the wind has a stationary distribution. You assume that the human perception of acceleration follows an unbiased Gaussian distribution. You also assume that a good model to describe the uncertainty in your initial guess of the ball's state is a Gaussian model, with the standard deviation given by your estimated bounds.

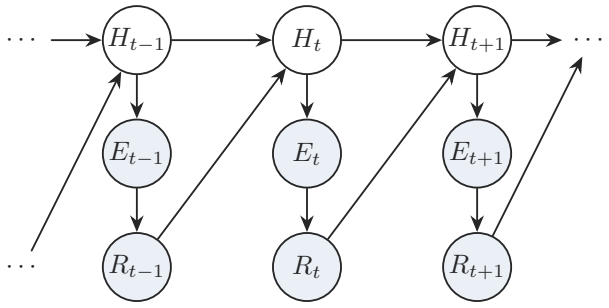
(a) You wish to estimate the current state of the ball given the perceptions of a participant. Define the components of a Kalman filter in this context.

(b) Express the distribution $p(\mathbf{x}_t \mid e_{1:t})$ with respect to the components defined previously. Use the cheat sheet for Gaussian distributions.

(c) Represent the transition and sensor models as a dynamic Bayesian network.

Exercise 6 Human-robot interaction (CS188, Fall 2017)

In the near future, autonomous robots will live among us. Therefore, the robots need to know how to act appropriately in the presence of humans. In this question, we explore a simplified model of this interaction. We assume that we can observe the robot's actions at time t, R_t, and an evidence observation, E_t, directly caused by the (hidden) human state, H_t. Robot actions from the current time-step affect the human state in the next-time step, as illustrated in the Bayesian network given below.



Assuming discrete random variables and given the probability tables $\mathbf{P}(H_0)$, $\mathbf{P}(E_t \mid H_t)$, $\mathbf{P}(R_t \mid E_t)$ and $\mathbf{P}(H_{t+1} \mid H_t, R_t)$, your goal is to derive a procedure to maintain a belief $\mathbf{P}(H_t \mid e_{1:t}, r_{1:t})$ about the state of the human at time t. At each time step, the robot's action r_t is observed after the evidence e_t.

(a) Derive the update step, which incorporates a pair of observations (e_t, r_t) into the predicted belief $\mathbf{P}(H_t \mid e_{1:t-1}, r_{1:t-1})$.

(b) Derive the predict step, which gives the predicted belief $\mathbf{P}(H_{t+1} \mid e_{1:t}, r_{1:t})$ about the state of the human at time $t+1$ from the belief $\mathbf{P}(H_t \mid e_{1:t}, r_{1:t})$.

(c) Combine both steps to derive a recursive update of the belief state $\mathbf{P}(H_t \mid e_{1:t}, r_{1:t})$, as observations are collected and time passes.

(d) Let us now assume that all variables are continuous. Discuss how you would compute or approximate the belief state on a computer.

Exercise 7 Server cooling (Exam, August 2019; AIMA ch. 14)
A monitoring program wants to know whether a server is cooled enough. Each day, the program checks whether the server raised a temperature alert, and whether it slowed down (throttled) its processors. The program makes the following hypotheses:

- The prior probability that the cooling is enough, with no observations, is 0.7.
- The probability that the cooling is enough on day t is 0.8 given that it was enough on the previous day, and 0.3 otherwise.
- The probability of a temperature alert is 0.2 if the cooling is enough, and 0.7 otherwise.
- The probability of throttling is 0.1 if the cooling is enough, and 0.3 otherwise.

(a) Formulate the environment and hypotheses as a dynamic Bayesian network that the program could use to detect insufficient cooling from a sequence of observations. Provide the associated probability tables.

(b) Reformulate the dynamic Bayesian network as a hidden Markov model that has only a single observation variable. Give the complete probability tables for the model.

(c) For the sequence $e_{1:3}$ of observations "no alert, no throttling", "alert, no throttling" and "alert, throttling", calculate the distributions $\mathbf{P}(C_t \mid e_{1:t})$ and $\mathbf{P}(C_t \mid e_{1:3})$ for $t \in \{1, 2, 3\}$.

Quiz

1. The normal distribution $\mathcal{N}(\mu, \sigma^2)$ is described by the density function
   - ◯ $p(x) = \frac{1}{\sigma} \exp\left(-(z + \exp(-z))\right)$, with $z = \frac{x-\mu}{\sigma}$.
   - ◯ $p(x) = \frac{1}{2\sigma} \exp\left(-\frac{|x-\mu|}{\sigma}\right)$.
   - ◯ $p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.
   - ◯ $p(x) = \dfrac{1}{\pi\sigma\left[1 + \left(\frac{x-\mu}{\sigma}\right)^2\right]}$.
2. The Kalman filter requires the specification of ...
   - ◯ a prior $p(\mathbf{x}_0)$ and a transition model $p(\mathbf{x}_{t+1} \mid \mathbf{x}_t)$.
   - ◯ a transition model $p(\mathbf{x}_{t+1} \mid \mathbf{x}_t)$ and a sensor model $p(\mathbf{e}_t \mid \mathbf{x}_t)$.
   - ◯ a prior $p(\mathbf{x}_0)$, a transition model $p(\mathbf{x}_{t+1} \mid \mathbf{x}_t)$, and a sensor model $p(\mathbf{e}_t \mid \mathbf{x}_t)$.
   - ◯ a prior $p(\mathbf{x}_0)$, a prior $p(\mathbf{e}_0)$, a transition model $p(\mathbf{x}_{t+1} \mid \mathbf{x}_t)$, and a sensor model $p(\mathbf{e}_t \mid \mathbf{x}_t)$.
3. In filtering, the belief state is updated at each timestep using ...
   - ◯ the current belief and the transition model.
   - ◯ the current belief, the sensor model and the new evidence.
   - ◯ the transition model, the sensor model and the new evidence.
   - ◯ the current belief, the transition model, the sensor model and the new evidence.

4. Let us assume a Markov process with hidden state variables $\mathbf{X}_t$, evidence variables $\mathbf{E}_t$, transition model $\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{X}_t)$ and sensor model $\mathbf{P}(\mathbf{E}_t \mid \mathbf{X}_t)$. Which of the following is true?

   ◯ The Bayes filter estimates $\mathbf{P}(\mathbf{X}_{1:t} \mid \mathbf{e}_{1:t})$.

   ◯ Smoothing consists of computing $\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t+1})$ by pushing the belief state $\mathbf{P}(\mathbf{X}_t \mid \mathbf{e}_{1:t})$ forward through the transition model and then updating this new belief state with the evidence $\mathbf{e}_{t+1}$.

   ◯ The Kalman filter computes the belief state exactly for non-linear transition models.

   ◯ The Viterbi algorithm computes the most likely sequence of state values $\mathbf{x}_{1:t}$ given the evidence $\mathbf{e}_{1:t}$.